

STUDENT COURSE PORTFOLIO

ITP111-System Administration and Maintenance

A.Y. 2025 – 2026 / 1st SEMESTER

SARINO, NHOEL IVAN A.

4IT-B





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(Pamantasan ng cabuyao)
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I. Vision/Mission of UC (PnC)

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OUTCOMES BASED TEACHING AND LEARNING PLAN in ITP111 – System Administration and Maintenance Department: CCS

VISION	CORE VALUES
An institution of higher learning in Region IV, developing globally-competitive and value-laden professionals and leaders instrumental to community development and nation building.	As a God-fearing institution respecting multi-faith of people, PnC adheres to the following core values: • Personal Dignity • Nurturing Community • Commitment to Excellence
MISSION	PnC (UC) GRADUATE ATTRIBUTES
An institution of higher learning committed to equip individuals with knowledge, skills and values that will enable them to achieve professional goals & provide leadership and service for national development.	1. Professional competent individual 2. Proficient communicator 3. Analytical & problem solver 4. Lifelong learner 5. Team leader or player 6. Productive individual 7. Socially and ethically responsible citizen 8. Technology-driven professional 9. Value-laden professional

II. PEOs of the Program

Three to five years after graduation, the Bachelor of Science in Information Technology alumni shall:

1. be employed in the fields of computing.
2. possess globally competitive skills.
3. demonstrate personal dignity and integrity; and
4. engage in community development or nation building.

III. SOs addressed by the course

SO1. Identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing-based systems.

SO2. Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession.

SO3. Understand best practices and standards and their applications.

SO4. Analyze the local and global impact of computing information technology on individuals, organizations and society.

SO5. Recognize professional responsibilities and make informed judgements in computing practice based on legal and ethical principles.



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IV. OBTLP or Courses Syllabi/Outline



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COURSE/SUBJECT CODE	ITP111	COURSE/SUBJECT TITLE	System Administration and Maintenance
CREDIT UNITS	3	COURSE/SUBJECT PRE-REQUISITE/CO-REQUISITE	ITP 105, ITP109
COURSE/SUBJECT DESCRIPTION	This course covers the installation and configuration of mainstream operating systems, important network services, disaster recovery procedures, and techniques maintenance and for ensuring the security of the system.		

TEACHING AND LEARNING PLAN

Week	Topics	Intended Learning Outcomes (ILO)	Teaching and Learning Activities (TLA)	Instructional Material	Assessment Tasks (AT)
1	Discussion of PnC Mission and Vision, PnC Core Values, PnC Graduate Attributes, Program Outcomes, Course Outcomes, Course Orientation and Policies. Introduction to System Administration - Role and responsibilities of a system administrator	1. Define the key responsibilities of a system administrator including user management, system configuration and security measures. 2. Describe the basic architecture and components of each operating system. 3. Define fundamental networking concepts, including IP addressing, subnetting and DNS resolution. 4. Learn how to create and manage virtual machines.	Lecture	Module	



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	- Operating System overview (Windows, Linux, macOS) - Networking fundamentals for system administrators - Introduction to virtualization technologies	Associated CIL04, CIL02			
2-3	Operating System Installation and Configuration - Planning and performing OS installations - Disk partitioning and file system management - Initial configuration and setup of operating systems - User and group management	1. Define the requirements for installing different operating systems. 2. Demonstrate how to partition disks and allocate storage space for different operating systems. 3. Identify how they manage user groups and the significance of group memberships. 4. Discuss the concept of system services and daemons and their roles in the operating system. 5. Discuss the feature of Network Operating System. 6. Demonstrate how to configure active directory.	Lecture Demonstration of VM setup and OS installation using VM	Module	Lecture: Quiz Seatwork Laboratory: Laboratory Exercises



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	- Managing system services and daemons	Associated CIL04, CIL03			
4-5	System Monitoring and Performance Tuning - Monitoring system performance using built-in tools - Analyzing system resource usage (CPU, memory, disk, network) - Identifying and troubleshooting performance bottlenecks - Performance tuning techniques	1. Recognize the used of built-in tools to monitor CPU usage and identify process consuming excessive resources. 2. Define the common performance bottlenecks that can impact system performance. 3. Explain various system configuration such as kernel parameters and system settings. Associated CIL01, CIL03	Lecture	Module	Lecture: Quiz Recitation Seatwork Case Study Laboratory: Laboratory Exercises
Prelim Examination					
7-8	System Backup and Recovery - Importance of system backups and disaster recovery planning	1. Discuss the significance of data backups and disaster recovery planning. 2. Learn how to create and schedule automated data backup using different backup types.	Lecture Demonstration of system backup and recovery failure	Module	Lecture: Seatwork Recitation Laboratory: Laboratory Exercises



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	- Implementing regular data backups and scheduling strategies - Utilizing backup tools and solutions - Recovering from system failures and data loss incidents	3. Familiarize with various backup software and built-in utilities to perform backup operations efficiently. 4. Demonstrate system recovery from data backups. 5. Discuss the significance of creating Preventive Maintenance Associated CIL02, CIL03, CIL04			
9-10	Security and Access Control - System security principles and best practices - User authentication methods (passwords, multi-factor authentication) - Implementing access control and permissions (ACLs, sudo) - System hardening techniques	1. Define different security threats and common attacks vectors. 2. Explain the concept of multi-factor authentication and its role in enhancing security. 3. Demonstrate how to configure file system permission and user/group access rights effectively. 4. Define the importance of regular and security audits and vulnerability assessments. Associated CIL0	Lecture	Module	Lecture: Seatwork Quiz Laboratory: Laboratory Exercises
11	Networking and Remote Administration	1. Define the basic IP addressing and subnetting.	Lecture	Module	Lecture: Quiz



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	- Configuration network interfaces and protocols - Managing networking services(DHCP, DNS, firewalls) - Remote administration tools (e.g. SSH, Remote Desktop) - Troubleshooting network connectivity issues	2. Demonstrate the configuration and maintenance of DHCP and DNS. 3. Define the security considerations on using remote administration. 4. Define how to interpret network error message and logs. 5. Demonstrate the configuration of file services, printer and document on the server. 6. Demonstrate how a client join to Server Domain. Associated CIL02, CIL03			Seatwork Case Study Laboratory: Laboratory Exercises
Midterm Examination					
13-14	Automation and Scripting - Introduction to scripting languages (Bash, PowerShell) - Automating system tasks with scripts - Configuration management tools(e.g. Ansible, Puppet)	1. Define the popular scripting languages like Bash and Powershell. 2. Demonstrate how to schedule scripts using Cron or Task Scheduler. 3. Recognize how to write and apply configuration management scripts. Associated CIL02, CIL03	Lecture Demonstration of using Powershell or Bash	Module	Lecture: Seatwork Laboratory: Laboratory Exercises



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15-16	Virtualization and Cloud Computing - Overview of virtualization technologies (e.g. VMware, Hyper-v) - Managing virtual machines and containers - Introduction to cloud computing and cloud-based services	1. Recognize popular virtualization technologies such as VMware, Hyper-V and others. 2. Demonstrate practical skills in creating, configuring, and managing virtual machines. 3. Recognize various cloud-based services such as cloud storage, computing resources and software applications. Associated CIL01, CIL03	Lecture	Module	Lecture: Quiz Seatwork Laboratory: Laboratory Exercises
17	Troubleshooting and Best Practices - Methodologies for systematic problem solving. - Diagnosing common system issues and errors. - Best practices for system administration and maintenance	1. Define different systematic troubleshooting methodologies such as troubleshooting triangle or divide and conquer. 2. Explain how to interpret error messages, logs, and system diagnostic tools to identify the source of problems. 3. Define industry best practices for system administration and maintenance. 4. Demonstrate the concept of disaster recovery planning and business continuity.	Lecture Documentation Projects Role-Play Scenario	Module	Lecture: Quiz Seatwork Case Study Laboratory: Laboratory Exercises



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	- Creating comprehensive system documentation - Developing disaster recovery plans and business continuity strategies.	5. Create comprehensive documentation for system configurations, network diagrams, and standard operating procedures (SOPs). Associated CILO1, CILO2, CILO3			
Final Examination					

TEXTBOOKS AND REFERENCE MATERIALS	Books: al., M.E. (2019). CSS and Networking Guide for CCNA and CSS. Unlimited Books Library Services & Publishing Inc. Barnes, R. (2020). Understanding Operating Systems. Lenrye International. Holcombe, J., & Holcombe, C. (2020). Survey of Operating System. McGraw Hill Education. Panek, C. (2020). Windows Operating System Fundamentals. Sybex. E-Book Andres, D. W. (2020). CompTIA A+ Core 2 Exam: Guide to Operating Systems and Security. Ciampa. (2022). CompTIA Security+ Guide to Network Security Fundamentals. West. (2021). CompTIA Cloud+ Guide to Cloud Co. Journal Luca Deri, A. C. (2022). Using CyberScore for Network Traffic Monitoring. IEEE. Mughal, A. A. (2022). Well-Architected Wireless Network Security. Journal of Humanities and Applied Science Research. Nikhil Aggarwal, R. R. (2022). Virtualization and Cloud Computing. IEEE.
GRADING SYSTEM	The final grade in this course will be composed of the following items and their weights in the final grade computation;



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COURSE REQUIREMENTS	Lecture (60%)		
	Assessment Item	Grade Source	Percentage
	Quizzes	Quiz	20%
	Class Participation	Recitation	10%
	Other Requirements	Research Project/Case Studies	20%
	Exam	Written Exam	50%
	Laboratory (40%)		
	Assessment Item	Grade Source	Percentage
	Quizzes	Laboratory Exercises	80%
	Exam	Laboratory Exam	20%
	Total		100%
CLASS POLICIES			

Attendance will be considered as a basis for dropping students. Below is the table showing the maximum number of absences against number of meetings per week.

# of meetings per week	Absences should not exceed (per sem)
5	17
3	10
2	7
1	3

Case Study
Portfolio

Class Attendance



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V. Answered Prelim Assessment Tasks (Lecture/Lab)



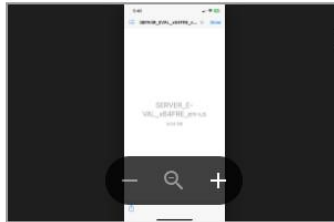
Download ISO File
Prelim Assignment

« Back to Class Page

Post Details

My Work

✓ Your work has been submitted.
August 8, 2025 @ 5:43 am



222.0 KB

Record ID #0100777-2201629-1300502



Lab Activity #1 & 2: Installation and Setup of Windows Server & Active Directory
Prelim Assignment

« Back to Class Page

Post Details

My Work

✓ Your work has been submitted.
September 1, 2025 @ 3:43 am

https://drive.google.com/drive/folders/1ZJyGchXYChcbFzvh1-ccOBH0y0TeDVAC?usp=drive_link

Your Grade

60 / 60
100.00 %

Record ID #0104337-2201629-1326868



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Prelim Lab Exam

Prelim Assignment

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Post Details

My Work

✓ Your work has been submitted.
September 20, 2025 @ 4:40 pm

https://drive.google.com/file/d/1kSKXX2wTzVh6dTL3OzaFYr4lIC_48qP7/view?usp=sharing

Your Grade

90 / 100
90.00 %

Record ID #0107106-2201629-1358328



ITP111 - Quiz #1

Prelim Online Quiz

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Post Details

My Work

Your Score

9 / 10
90.00 %

Started: August 16, 2025 @ 4:15 PM

Finished: August 16, 2025 @ 4:18 PM

Record ID #0102077-2201629-1305827



Quiz #2: Operating System Installation and Configuration

Prelim Online Quiz

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Post Details

My Work

Your Score

9 / 10
90.00 %

Started: September 1, 2025 @ 2:44 PM

Finished: September 1, 2025 @ 2:47 PM

Record ID #0104336-2201629-1328003



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VI. Learning Reflection

This term introduced me to the core duties of a system administrator. I learned the importance of user management, system configuration, and security measures. The most useful activity was installing and setting up the Windows Server Operating System and Active Directory. Through the laboratory exercises, I gained practical skills in creating virtual machines and partitioning disks. These tasks helped me understand the technical requirements for building a stable server environment.

VII. Answered Midterm Assessment Tasks (Lecture/Lab)

Week 7-8 Activity
Midterm Assignment

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Post Details

My Work

✓ Your work has been submitted.

October 26, 2025 @ 11:50 pm

MendozaSarino_BackupRecoveryLab-1.pdf

805.3 KB

Record ID #0111389-2201629-1425638

Week 9-10 Activity
Midterm Assignment

« Back to Class Page

Post Details

My Work

✓ Your work has been submitted.

October 21, 2025 @ 11:59 pm

WEEK 9-10 (Mendoza, Miranda, Refrea, Sarino, Soriano).pdf

443.7 KB

Record ID #0111392-2201629-1413898



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Midterm Lab Exam
Midterm Assignment

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Post Details

My Work

✓ Your work has been submitted.
November 16, 2025 @ 2:59 pm

Midterm_Exam_Sarino.pdf

1.5 MB

Your Grade

76 / 100

76.00 %

Instructor's Comment

No recommendation. No windows backup and recovery. Images must have description.

Record ID #0114157-2201629-1441397



Midterm Quiz 5
Midterm Online Quiz

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Post Details

My Work

Your Score

19 / 20

95.00 %

Started: November 17, 2025 @ 2:41 PM

Finished: November 17, 2025 @ 2:44 PM

Record ID #0114164-2201629-1442405



Midterm Quiz 6
Midterm Online Quiz

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Post Details

My Work

Your Score

15 / 16

93.75 %

Started: November 17, 2025 @ 2:39 PM

Finished: November 17, 2025 @ 2:40 PM

Record ID #0114165-2201629-1442404



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 **Midterm Activity 5**
Midterm Online Quiz

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Post DetailsMy Work

Your Score

10 / 10

100.00 %

Started: November 17, 2025 @ 2:36 PMFinished: November 17, 2025 @ 2:38 PM

Record ID #0114166-2201629-1442400

VIII. Learning Reflection

The focus of this term was on maintaining system health and security. I studied how to monitor system performance and the critical need for data backups and disaster recovery planning. The laboratory activities regarding system backup were very helpful in understanding how to protect data. I also learned about implementing access controls to secure the system against threats. The feedback from my exams taught me that proper documentation is just as important as technical skills when recovering systems.

IX. Answered Finals Assessment Tasks (Lecture/Lab)

 **Finals Quiz 1 & 2**
Finals Online Quiz

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Post DetailsMy Work

Your Score

19 / 20

95.00 %

Started: October 25, 2025 @ 1:26 PMFinished: October 25, 2025 @ 1:33 PM

Record ID #0112958-2201629-1422455



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X. Learning Reflection

The final part of the course covered advanced topics like networking and automation. I learned how to manage network services such as DHCP and DNS, which are essential for server communication. The lessons on scripting languages like Bash and PowerShell showed me how to automate repetitive tasks to be more efficient. Additionally, studying virtualization and cloud computing gave me a better understanding of modern IT infrastructure. These lessons expanded my ability to handle complex administrative tasks.

XI. Overall Course Learning Reflection

This course has provided me with essential skills for the IT profession. I have progressed from understanding basic operating system installation to managing complex security configurations and disaster recovery plans. The combination of lectures and hands-on laboratory activities allowed me to apply theoretical knowledge to real-world scenarios. I now fully understand the professional responsibilities of a system administrator in ensuring systems are reliable and secure. I am confident that the technical and analytical skills gained here will be valuable for my future career.

XII. Suggestions/Recommendations

I do not have any specific suggestions or recommendations.